



Review

Toxicity of polybrominated diphenyl ethers (PBDEs) on rodent male reproductive system: A systematic review and meta-analysis of randomized control studies

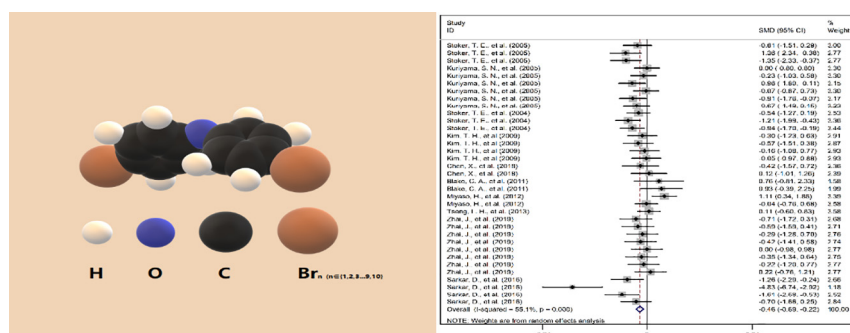
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HIGHLIGHTS

- This was the first meta-analysis to illuminate the relationship between PBDEs and male reproductive system.
- PBDEs exposure may damage the development of the male reproductive system.
- High dosage of PBDEs gave a more significant adverse effect on male reproductive health.
- The pregnant and the infant should be kept away from PBDEs.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 28 October 2019
Received in revised form 14 February 2020
Accepted 17 February 2020
Available online 19 February 2020

Editor: Daqiang Yin

Keywords:

PBDE
Male reproductive
Meta-analysis
Rodent

ABSTRACT

Polybrominated diphenyl ethers (PBDEs) are a group of environmental endocrine-disrupting chemicals (EDCs). According to recent studies, the associations between PBDEs and rodent male reproductive system are controversial. Therefore, we performed this systematic review and meta-analysis to clarify the effects of PBDEs on rodent male reproductive system. Data were extracted from articles collected from PubMed, Web of Science, and other databases. The pooled standard mean deviation (SMD) with 95% confidence interval (95% CI) was used to evaluate the association between the male rodent reproductive system impairment and PBDE exposure. Ten articles were included in the present meta-analysis and systematic review. The summary SMD of reproductive toxicity associated with PBDEs was -0.46 (95% CI: -0.69 , -0.22), and the overall association between DE-71 along with BDE-209 and male reproductive system damage was statistically significant (SMD = -0.7 95% CI: -1.18 , -0.21 , $p = .041$; SMD = -0.41 , 95% CI: -0.77 , -0.05 , $p = .000$). The adverse impact of PBDEs on rodent male reproductive system, especially seminal vesicle (SMD = -1.09 , 95% CI: -1.49 , -0.49 , $p = .523$) and ventral prostate (SMD = -1.27 , 95% CI: -1.88 , -0.65 , $p = .821$), were clearly demonstrated. Moderate (SMD = -0.81 , 95% CI: -1.182 , -0.437 , $p = .197$) and high (SMD = -0.41 , 95% CI: -0.76 , -0.05 , $p = .000$) dosage may have exerted effect than relatively low (SMD = -0.29 , 95% CI: -0.17 , 0.12 , $p = .136$) dosage. In conclusion, our meta-analysis and systematic review suggested PBDE toxicity on the male reproductive system.

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1. Introduction

Environmental safety has been a serious concern for many years. Numerous different chemicals exist in our environment, and some of which are known to have adverse effects on the homeostasis or development of the endocrine, cardiovascular, reproductive, and neurological systems in humans and other species. These chemicals are often classified as endocrine-disrupting chemicals (EDCs), such as various types of pesticides, plasticizer, and certain aromatic compounds (Wang et al., 2016). A new group of chemicals with similar structure called polybrominated diphenyl ethers (PBDEs) are classified as EDCs (Eskenazi et al., 2017).

The molecular structure of PBDEs contains two phenol rings connected by an oxygen atom, with a number of bromine atoms binding on the two phenol rings. Based on the various locations and numbers (1–10) of the bromine substituent, 209 different PBDE congeners and some commercial mixtures, such as like DE-71, are created (Lilienthal et al., 2006). PBDEs are widely used as brominated flame retardants in electronics, textile products, and commodities containing polyurethane (Meeker et al., 2009). Consequently, PBDEs spread widely in the environment from electronic equipment, furnitures, and construction materials (Turyk et al., 2008). In the USA, PBDEs released from the disposal of electronic and electrical equipment (EEE) waste, which accounted for approximately 2.4 million metric tons (MMT) of EEE waste produced in 2005, raises widespread concern. Furthermore, 1.68 MMT of these 2.4 MMT EEE wastes were merely buried in municipal solid waste landfills. Therefore, landfill sites have become reservoirs of PBDEs, polluting the soil and water reservoirs of landfill sites and the surrounding environment. Eventually, PBDEs accumulate in different animals through the food chain and had been detected in birds, mammals, fish, eggs, and even human (Turyk et al., 2008; EPA, 2010; Labunska et al., 2013).

The toxicity of PBDEs on different organs in different systems have been verified in several studies (Dorman et al., 2018; Lee et al., 2010; Zhao et al., 2011; Van der Ven et al., 2008). PBDEs are associated with high morbidity of hypothyroidism in humans (Oulhote et al., 2016) and neurotoxicity to animals (Dorman et al., 2018). Whether PBDEs have adverse effects on the male reproductive system remains inconclusive. Several human studies suggested that severe cryptorchidism (Goodyer et al., 2017) and poor sperm quality (Yu et al., 2018) are significantly associated with elevated PBDE levels in maternal serum, hair, and even house dust (Johnson et al., 2013). For rodent species, even a single gavage dose of 300 or 60 µg/kg of PBDE-99 on the 9th gestational day could significantly decrease rats' spermatogenesis and relative epididymis weight (Kuriyama et al., 2005). Meanwhile, a dose of 60 or 120 mg/kg DE-71 exposed from postnatal day (PND) 23 to PND 53 distinctly decreased the seminal vesicle weight and ventral prostate weight (Stoker et al., 2004; Stoker et al., 2005). However, numerous

studies did not find an impact by high dose of PBDE on several organs or serum marks, and the effects of PBDE exposure on male reproductive system remain inconsistent (Tseng et al., 2013; Kim et al., 2009; Blake et al., 2011).

Accumulating evidence from observations of epidemiological studies and individual patients has shown an increase in the incidence of testicular neoplasm, reduced seminal quality, genital abnormalities, and weak fertility or infertility. Environmental factors, such as EDCs, may have potential adverse effects that deserve attention (Skakkebaek et al., 2001). Direct exposure to food and dust contaminated by EDCs can damage an individual's male reproductive system, such as declined serum testosterone concentration (Johnson et al., 2013), and, in worst cases, indirect maternal exposure during antenatal time may also impair the male reproductive health of offsprings (Garcia-Villarino et al., 2018; Luan et al., 2019). Moreover, male reproductive dysfunction may act as a predictor of depression disorders associated with suicidal rate (Kennedy et al., 2011). To manage these chemicals and protect male reproductive health, we need to explain clearly the potential reproductive toxicity of these chemicals.

2. Methods

2.1. Literature search

Our research design was generally developed according to (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) PRISMA Statement (Moher et al., 2009a; Moher et al., 2009b). The topic keywords were established according to the PICO principle (participants, interventions, comparisons, and outcomes): P as rodent species; I as PBDE exposure; C as control groups; and O as male reproductive system outcomes. The topic statement of our study question was developed clearly as “Do PBDEs have adverse effects on rodent male reproductive system?” According to our topic keywords, we first searched PubMed, Cochrane Library, and Web of Science databases up to September 2019 and generally used the following keywords to identify available studies: “semen”, “spermatogenesis”, “spermatocyte”, “infertility”, “seminal”, “testicular”, “reproductive”, “cryptorchidism”, “hypospadias”, “Leydig”, “gonocytes”, “testosterone”, “polybrominated diphenyl ether”, “PBDE”, “halogenated”, “brominated”, “DE”, “bromo”, “BDE”, “mouse”, “mice”, “rat”, and “rats”. The references of included studies were also browsed to manually obtain several articles. All searched articles were initially screened by reading their abstracts. The full texts of preliminarily included articles were re-screened to decide the final eligible articles. The language of included articles was limited to English, and only rodent studies could be included. A flow diagram detailing the selection of studies is shown in Fig. 1.

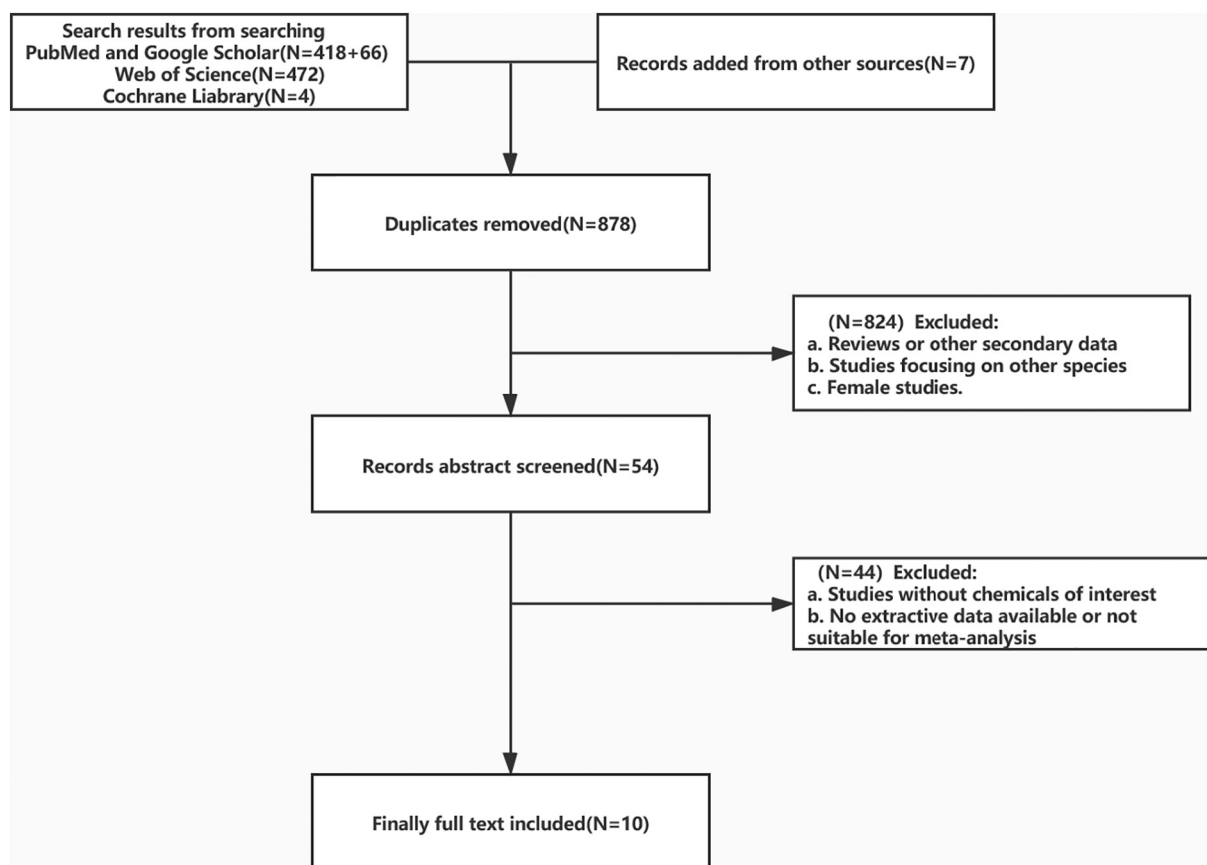


Fig. 1. Flow chart of study search progress.

2.2. Inclusion and exclusion criteria

Studies meeting the following criteria were included in the present meta-analysis: (1) referred to the effects of PBDEs on the reproductive system of rodent species and (2) rodent randomized control trials (RCTs).

Studies meeting the following criteria were excluded: (1) non-original studies, such as editorials, reviews, and comments; (2) female studies; and (3) data types, which could not be used in the meta-analysis. Additionally, when the same sample species was investigated or mixed in different studies, only the most recent or most complete studies were included.

2.3. Data extraction and meta-analysis

Two independent reviewers extracted the data from each included article. The data extracted from the eligible studies were recorded as follows: author, year of publication, study type, chemical species, life stage of exposure, life stage of assessment, manner of exposure, dosage, and general results about the changes in the reproductive system.

The heterogeneity between the included studies was quantified by Chi-square test and I-square test and was considered to be obvious if p was <0.10 or I^2 was $>50\%$. The mean and standard deviation of the maximum dose groups of the included articles were synthesized to calculate the standardized mean difference (SMD). When p was <0.10 or I^2 was $>50\%$ (high heterogeneity), random-effects models were applied. When p was >0.10 and I^2 was $<50\%$ (low heterogeneity), fixed-effects models were used. Stata software (version 12.0; Stata Corp. LP, College Station, TX) was used for the present meta-analysis. Moreover, Begg's test was utilized to evaluate the publication bias among the studies. Two-tailed p values <0.05 were considered statistically significant.

3. Results

3.1. Study characteristics

Ten articles were finally included in the present meta-analysis and systematic review. All eligible studies were all toxicological RCTs on rodent species (rat or mouse) associated with PBDEs, which covered deca-BDE (also called BDE-209), PBDE-99 and BDE-3, and a common PBDE mixture called DE-71. Among these studies, six studies on epididymis weight, five articles on testes weight, six investigations on sperm number, and three studies on testosterone concentration were included (Kuriyama et al., 2005; Stoker et al., 2004; Stoker et al., 2005; Tseng et al., 2013; Kim et al., 2009; Blake et al., 2011; Zhai et al., 2019; Chen et al., 2018; Miyaso et al., 2012; Sarkar et al., 2016). All articles were published between 2000 and 2019 (Kuriyama et al., 2005; Stoker et al., 2004; Stoker et al., 2005; Kim et al., 2009). Three studies were from America (Stoker et al., 2004; Stoker et al., 2005; Blake et al., 2011), three studies from China (Tseng et al., 2013; Zhai et al., 2019; Chen et al., 2018), one study from Japan (Miyaso et al., 2012), one study from Germany (Kuriyama et al., 2005), one study from India (Sarkar et al., 2016), and one study from Korea (Kim et al., 2009). Detailed characteristics of the included studies are summarized in Table 1.

3.2. Toxicity of PBDE on rodent male reproductive system

Ten studies contributing to 35 SMD estimators were analyzed. The summary SMD of toxicity of PBDEs on rodent male reproductive system was -0.46 (95% CI: -0.69 , -0.22) according to the random-effects model (Fig. 2). In addition, sub-group analyses individually based on specific tissues, chemical class and dosage, and exposure life stage were performed. The overall association between DE-71 along with BDE-209 and male reproductive system damage was statistically

Table 1
Main data characteristics of the included studies.

Study	Chemical	Study type	Species	N in analysis	Life stage exposed	Life stage assessed	Method	Dose	Results
Stoker et al. (2004)	DE-71	radomized control trial	Wistar rats	15	PND23-PND53	PND53	gavage	0,3,30,60 (mg/kg)	Serum Testosterone (ng/ml)
Stoker et al. (2005)	DE-71	radomized control trial	Wistar rats	10	PND23-PND53	PND53	gavage	0,60,120 (mg/kg)	Epididymis(g)
Kuriyama et al. (2005)	PBDE-99	radomized control trial	Wistar rats	12	GD6	PND140	gavage	0,60,300(μg/kg)	Epididymis(g) Testes(g) Sperm number (10 ⁶ /ml) Serum Testosterone (ng/ml) Testes/bw(mg/g) Epididymis/bw (mg/g)
Kim et al. (2009)	BDE-209	radomized control trial	SD rats	4	G6-G18	PND42	gavage	0,5,40,320 (mg/kg)	Epididymis(g) Testes(g) Testes/bw(mg/g) Epididymis/bw (mg/g)
Blake et al. (2011)	DE-71	radomized control trial	CD@IGS rats	3 ∙ 5	G1.5-PND21 (except PND1)	PND21 ∙ PND150	gavage	0,60(μg/kg)	Testes/bw(mg/g)
Miyaso et al. (2012)	deca-BDE	radomized control trial	ICR mice	15	PND1-PND5	PND84	subcutaneous injection	0,0.025,0.25,2.5 (mg/kg)	Epididymis(g) Testes(g)
Tseng et al. (2013)	BDE-209	radomized control trial	CD-1 mice	15	GD0-GD17	PND71	gavage	0,10,500,1500 (mg/kg)	Sperm number (10 ⁶ /ml)
Sarkar et al. (2016)	BDE-209	radomized control trial	Parkes strain male mice	9	totally 35 days	1 day after the last dosage	gavage	0,750,950 (mg/kg)	Sperm number (10 ⁶ /ml) Serum Testosterone (ng/ml) Testes/bw(mg/g) Epididymis/bw (mg/g)
Chen et al. (2018)	BDE-3	radomized control trial	SD rats	6	PND35-PND56	PND56	gavage	0,50,100,200 (mg/kg)	Epididymis(g) Testes(g)
Zhai et al. (2019)	BDE-209	radomized control trial	ICR mice	8	GD7-PND21	PND35 ∙ PND105	gavage	0,100,300,500 (mg/kg)	Epididymis(g) Testes(g) Testes/bw(mg/g) Epididymis/bw (mg/g)

G or GD: day after pregnancy, PND: post-natal day, VP: ventral prostate, LLP: Left lateral prostate, RLP: Right lateral prostate, EPI: epididymis, SV: seminal vesicle.

significant ($SMD = -0.7$ 95% CI: $-1.18, -0.21, p = .041$; $SMD = -0.41$, 95%CI: $-0.77, -0.05, p = .000$). In the sub-group analysis of specific tissues in reproductive system, the association between a decreased weight of ventral prostate along with seminal vesicle and PBDE exposure was considered significant ($SMD = -1.27$, 95% CI: $-1.88, -0.65, p = .821$; $SMD = -1.09$, 95% CI: $-1.49, -0.49, p = .523$). Dosage was divided into three groups: low (<1 mg/kg), moderate (1–200 mg/kg), and high groups (>200 mg/kg). Moderate ($SMD = -0.81$, 95% CI: $-1.182, -0.437, p = .197$) and high ($SMD = -0.41$, 95% CI: $-0.76, -0.05, p = .000$) PBDE dosages seemed to show more significant adverse effects than the low group ($SMD = -0.29$, 95% CI: $-0.17, 0.12, p = .136$). We also found that exposure at the antenatal time ($SMD = -0.31$, 95% CI: $-0.55, -0.07, p = .743$) and postnatal time ($SMD = -0.95$, 95% CI: $-1.51, -0.38, p = .000$) exerted the toxicity effect than that at the perinatal time ($SMD = -0.17$, 95% CI: $-0.5, 0.17, p = .645$). The results of all sub-group analyses are shown in Table 2.

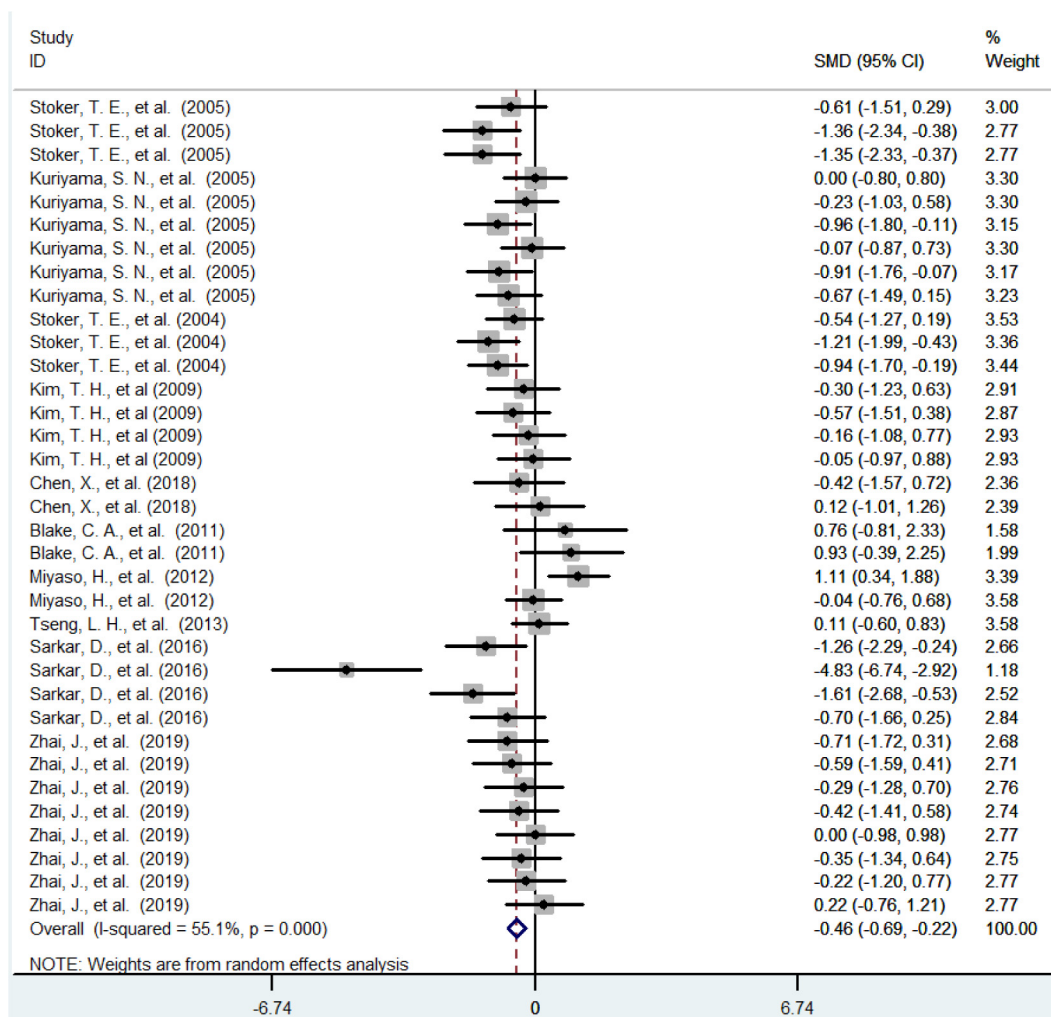
3.3. Test of heterogeneity and publication bias analysis

The pooled SMD of the association between PBDE exposure and male reproductive system impairment was conducted by applying the random-effects model. Begg's funnel plot was used to evaluate the publication bias of the studies, and the plot showed no evidence of obvious

asymmetrical shape, suggesting no significant publication bias (Fig. 3). In general, the outcome of our meta-analysis was acceptable.

4. Discussion

PBDEs are widespread environmental pollutants recognized during the past few years. PBDEs are used as fire retardants in different products and have received increasing attention due to their persistence, bioaccumulation, and potential harmful effects on humans and other species (Eguchi et al., 2016). The fractions of these compounds may escape from products when producing, using, disposing, and recycling of products containing PBDEs when added in products not chemically bound (e.g., covalently bound) but by physical combination (Meeker et al., 2009; Linares et al., 2015). As a result, human and other species are affected through food intake, drinking water, and touching dust from indoors and outdoors (EPA, 2010). The adverse effects of PBDEs on development outcomes, neurotoxicity, thyroid dysfunction, and potential carcinogenesis are recently detected (Dorman et al., 2018; Kuriyama et al., 2005; Buttke et al., 2013; Kuriyama et al., 2007). In male reproductive studies, sperm count, sperm motility, alterations of sperm chromatin, organ weight, and androgen-related compound concentration are the most common indicators examined to evaluate the extent of damage. Alterations in these parameters may indicate reproductive system damage. Many animal studies have provided evidence of male reproductive toxicity of PBDEs (Stoker et al., 2005; Tseng



SMD: standard mean deviation; 95%CI: 95% confidence intervals

Fig. 2. Forest plots for the effects of PBDEs on rodent reproductive system.

et al., 2013; Kim et al., 2009; Wang et al., 2008). Kuriyama et al. found that low doses of BDE-99 during development period significantly reduced sperm and spermatid counts (Kuriyama et al., 2005). In addition, PBDE intake may be associated with puberty retardation and decreased seminal vesicle and ventral prostate weights (Stoker et al., 2004; Stoker et al., 2005). Epidemiology studies also indicated associations between PBDE exposure and male reproductive diseases. A Swedish case-control study that included 44 cases of testicular malignant tumors found that the risk of testicular malignant tumors significantly increased (odds ratio = 2.5, 95% CI: 1.02, 6.0) associated with serum PBDE concentration (Hardell et al., 2006). In Canada, 137 cases of cryptorchidism were included in a case-control study. Logistic regression analysis showed that the PBDE-99-100-154 concentration in the hair of pregnant women was significantly correlated with the severity of cryptorchidism (Goodyer et al., 2017). However, statistical significance was not detected between neonatal hypospadias and maternal serum PBDE levels. The association between testicular cancer incidence and prenatal PBDE exposure was also insignificant in another study (Goodyer et al., 2017; Carmichael et al., 2010).

The definite impact of PBDEs on male reproductive system remains inconclusive according to previous studies with varying results (Goodyer et al., 2017; Hardell et al., 2006). Accepting any conclusion of individual studies may risk overgeneralization due to the limitations of region, sample size, and study design. Meta-analysis and systematic

review are a kind of statistical analysis combining the results from multiple scientific studies that use statistical approaches to develop pooled statistical values to estimate the unknown general common truth. In meta-analysis, the conflicting results from different studies are weighted and quantified on the basis of several factors, such as reliability, sample size, heterogeneity, and study design. Moreover, data from different environmental pollutant-associated studies can be synthesized into a single analysis, wherein sample size is the summation of all studies included, thereby greatly improving the reliability of the analytical results compared with the results of single studies. Meta-analysis and systematic review are a reliable tool to explain whether a certain factor is associated with several clinical phenomena and are widely used in environmental studies.

To the best of our knowledge, the current meta-analysis is the first study on the effect of PBDE exposure on reproductive system. In our meta-analysis, clear association was observed between PBDE exposure and the damage on male reproductive system (SMD = -0.46, 95% CI: -0.69, -0.22). This finding was consistent with the result concluded by numerous studies that PBDE exposure causes reproductive endocrine system damage (Abdelouhab et al., 2011; Makey et al., 2016; Albert et al., 2018). BDE-209 and DE-71 may be more toxic than the other PBDEs included. Ventral prostate and seminal vesicle weights were sensitive indicators according to the sub-group analysis we performed. Stoker et al. reported the anti-androgenic effect of DE-71 and

Table 2

Detailed results of sub-group analysis based on chemical class, specific tissue, life stage of exposure and dosage.

Analysis	SMD (95%CI)		model	Heterogeneity	
				I ² (%)	P _{het}
Subgroup 1					
Chemical class					
DE-71	−0.70	−1.18, −0.21	Random	52.2	0.040
PBDE-99	−0.45	−0.79, −0.21	Random	3.6	0.394
BDE-209	−0.41	−0.77, −0.55	Random	63.3	0.000
Subgroup 2					
Specific tissue					
Epididymis(g)	−0.09	−0.59, 0.42	Random	51.9	0.052
Ventral prostate(g)	−1.27	−1.88, −0.55	Random	0.0	0.812
Seminal vesicle(g)	−1.09	−1.69, −0.49	Random	0.0	0.532
Testes(g)	−0.26	−0.63, 0.11	Random	0.0	0.894
Sperm number(10 ⁶ /ml)	−0.65	−1.51, 0.21	Random	66.9	0.049
Serum testosterone (ng/ml)	−0.55	−3.4, 0.31	Random	90.2	0.0
Testes/bw(mg/g)	−0.32	−0.9, 0.27	Random	52.6	0.049
Epididymis/bw(mg/g)	−0.35	−0.76, 0.07	Random	0.0	0.586
Subgroup 3					
Life stage of exposure					
Antenatal	−0.31	−0.55, −0.07	Random	0.0	0.743
Perinatal	−0.17	−0.5, 0.17	Random	0.0	0.645
Postnatal	−0.95	−1.51, −0.38	Random	78.8	0.000
Subgroup 3					
Dosage					
Low(0-1 mg/kg)	−0.29	−0.70, 0.12	Random	36.7	0.136
Moderate(1-200 mg/kg)	−0.81	−1.18, −0.44	Random	30.3	0.197
High(>200 mg/kg)	−0.46	−0.69, −0.22	Random	61.3	0.000

SMD: standard mean deviation; 95%CI: 95%confidence intervals; P_{het}: p value for heterogeneity.

found that seminal vesicle and ventral prostate weights decreased with a dose–response effect, including in low-dosage groups (Stoker et al., 2005). In their study, ventral prostate and seminal vesicle weights decreased consistently, which supported our results (Stoker et al., 2004). Testicular homeostasis disturbances of PBDEs were also detected. Exposure during postnatal life stage to low-dose BDE-209 (0.025 mg/kg) could inhibit mouse testicular development, decrease sperm counts and Sertoli cell numbers, suppress testicular weight and elongated spermatid number by increasing Tyr phosphorylation of CTTN (Miyaso et al.,

2012). BDE-209 increased H₂O₂ production in sperm, facilitating oxidative stress, thereby causing severe sperm DNA damage as reflected by sperm head abnormality (Tseng et al., 2013). DE-71 exposure improved hepatic CYP2B, CYP1A1, and T4-UDGPT activity and then increased the degradation of thyroid hormones. DE-71 exposure may be another reason for the impairment of testicular homeostasis by downregulating the expression of SF-1, leading to spermatogenesis suppression (Lee et al., 2010; Stoker et al., 2004; Sarkar et al., 2016). A disruption of tight junction proteins and basal ectoplasmic specialization functions and increased estrogen receptor α signaling pathway observed by Zhai et al. also possibly contribute to spermatogenesis defects (Zhai et al., 2019). The testicular weight of mice significantly decreased accompanied by the degenerative changes in the seminiferous tubules when treated by 950 mg/kg dose of BDE-209, and these toxicological changes seemed to be dose responsive (Sarkar et al., 2016). The mechanisms for each congener to affect the reproductive endocrine system may vary. For different PBDE congeners, the effects are not always the same. The dominant PBDE congener in the human body is BDE-47 (15.9%) (Chao et al., 2010). BDE-47 exposure in vitro increased testosterone production with a dose-dependent model by stimulating StAR and p450cc and was observed to have a stimulatory action of testosterone secretion; however, BDE-99, BDE-3, and BDE-209 acted differently from this situation, decreasing serum testosterone concentration (Zhao et al., 2011; Chen et al., 2018; Sarkar et al., 2016; Maranghi et al., 2013). Therefore, although PBDEs have an overall damaging effect on the male reproductive system, conclusions cannot be made when it comes to each specific compound.

Scientists must urgently conduct research to develop protective drugs against PBDEs to further understand its reproductive toxicity and formulate countermeasures. Moreover, further toxicological mechanism research is warranted. The mechanism of PBDE-induced reproductive system function and structural damage has been described in the previous section. However, the study on protective drugs is obviously insufficient. Bisphenol-A (BPA) is a widely used plasticizer and also an environmental endocrine disruptor like PBDEs. Studies have demonstrated that BPA compounds can damage the male reproductive system through oxidative stress, and the antioxidant vitamin C and vitamin A can play an important role in protecting against oxidative effects (Aikawa et al., 2004; Chitra et al., 2003). As mentioned above, oxidative stress is an important route of PBDE-induced reproductive system damage, and studies have found that exposure to PBDE affects the balance of vitamin A concentration in the body (Ellis-Hutchings et al., 2006). Thus, vitamins may be used as blocking drugs against PBDE reproductive toxicity, possibly contributing to protecting humans exposed to the atmosphere of PBDEs (Cullum and Zile, 1985). In addition to vitamins, certain plant extracts exhibit antioxidant effects. Studies have reported that Troxerutin can alleviate the damaging effects of PBDE in in vitro tests, but its in vivo tests for PBDE reproductive toxicity are incomplete (Qin et al., 2019; Shan et al., 2018; Zhang et al., 2015). In summary, research on protective drugs to block PBDE-induced damage to male reproductive system is insufficient and requires attention.

Although the male reproductive toxicity of PBDEs is generally estimated by our present meta-analysis, several limitations of this review cannot be ignored. Studies on this issue were complicated because of their individual study designs, especially the time of exposure. Among the studies in the present review, the time span of exposure started at GDO (Tseng et al., 2013) and lasted to PND56 in another study. Therefore, identifying which is the most sensitive damage phase of reproductive system when exposed to PBDEs is difficult (Chen et al., 2018). In addition, articles containing the same outcome indicators with outcomes displayed not numerically were few. Although no significant publication bias was detected, the heterogeneity between each study should be taken carefully. In addition, potential bias still cannot be ignored in view of the test efficiency of such tests, although Egger's test for publication bias did not show significant results. Further animal and human studies of high quality are expected to be analyzed to

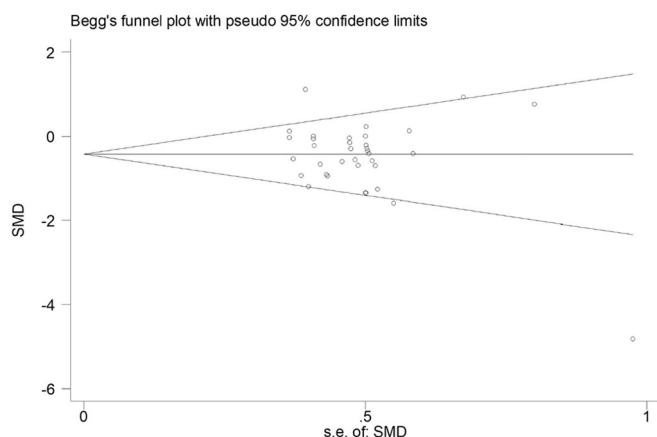


Fig. 3. Funnel plot of all studies included (Begg's test, $p = .320$). No significant publication bias was detected.

support our result. We look forward to conducting further research on the mechanisms to enrich the relevant content.

5. Conclusion

In summary, the results of our systematic review and meta-analysis provide evidence that PBDEs may impair the male reproductive system. Additional studies with detailed data are warranted to further explain the relationship between the adverse effects of PBDE exposure and male reproductive health. The results of our meta-analysis are essential for environmental study and suggest that pregnant women and infants should avoid PBDE exposure. In addition, associated safeguard procedures may need to be formulated by toxicologists.

Author contributions

ShangQian Wang and Zengjun Wang: Conceptualization, methodology, software, and supervision.

Tongtong Zhang, Xiang Zhou, and Aiming Xu: Data extraction, writing, and original draft preparation.

Ye Tian and Yuhao Wang: Reviewing and validation.

Yao Zhang and Qi Gu: Editing.

Funding information

This work was supported by the National Natural Science Foundation of China (No. 81800587).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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